

Name: KEY

Math 3108 101– Linear Algebra
Exam 1
September 13, 2024
Brigham

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned off or completely silenced (i.e. not on vibrate) for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 4) Failure to follow directions specific to a problem will result in the loss of points.
- 5) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 6) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 7) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 8) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos\pi$ should be calculated.
- 9) If you need extra room, you may use the back of the previous page. However, you must indicate you are doing so by clearly writing “BPP” on the relevant problem.
- 10) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 11) Once this exam begins, you will have 50 minutes to complete your solutions.

Page	1	2	3	4	5	Sum
Points Earned						
Maximum Points	18	20	20	21	21	100

DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO

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1. (10 points) Consider the matrix $B = \begin{bmatrix} 1 & 5 & -1 & -2 & 1 \\ 0 & 0 & 1 & -3 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

a) B is in _____.

(circle one)

Echelon Form

Reduced Echelon Form

Neither

b) The transformation $T(\mathbf{x}) = B\mathbf{x}$ is _____ and _____.

(circle one)

One-to-One

Not One-to-One

(circle one)

Onto

Not Onto

c) The columns of B span $\mathbb{R}^3$.

(circle one)

True

False

d) Fill in the blanks: If $T(\mathbf{x}) = B\mathbf{x}$ then $T: \mathbb{R}^5 \rightarrow \mathbb{R}^3$ (Domain and Codomain)

2. (2 points) Fill in the blank. The _____ of the transformation $T(\mathbf{x}) = A\mathbf{x}$ is the set of all images $T(\mathbf{x})$.

a) domain

b) range

c) codomain

d) none of these

3. (2 points) If $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^4$ such that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ form a linearly independent set, then $\mathbf{v}_4$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$.

(circle one)

True

False

4. (2 points) If A is an $m \times n$ matrix with m pivots, then $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

(circle one)

True

False

5. (2 points) If A is an $m \times n$ matrix with m pivots, then $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b} \in \mathbb{R}^m$.

(circle one)

True

False

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Open Answer. Show ALL your work!

6. (4 points each) Let $A = \begin{bmatrix} -2 & 3 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 2 & 1 \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$. Compute each of the following quantities, if it exists. If it does not exist, clearly explain why.

a) AB $\begin{bmatrix} -2 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 6 & 9 \\ 0 & 2 & 1 \end{bmatrix}$

b) $\mathbf{x}^T B^T$ $\begin{bmatrix} -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ -3 & 1 \end{bmatrix}$ DNE
of columns of $\mathbf{x}^T \neq$
of rows of B^T

c) $B\mathbf{x}$ DNE # of columns of $B \neq$ # of rows of $\mathbf{x}$

7. Let $A = \begin{bmatrix} 5 & -2 \\ 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$, and $C = \begin{bmatrix} 2 & -3 \\ -4 & 6 \end{bmatrix}$.

- a) (4 points) Compute the sum $A + C$.

$$\begin{bmatrix} 7 & -5 \\ -1 & 7 \end{bmatrix}$$

- b) (4 points) Compute the difference $C - B$.

$$\begin{bmatrix} 0 & -7 \\ -4 & 3 \end{bmatrix}$$

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8. (14 points) Solve the matrix equation $A\mathbf{x} = \mathbf{b}$ using augmented matrices, where

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \\ 3 & 7 & -1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} -2 \\ 2 \\ -4 \end{bmatrix}.$$

Show all your work! State your solution in parametric vector form.

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & -2 \\ 0 & 1 & 5 & 2 \\ 3 & 7 & -1 & -4 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 4 & -2 \\ 0 & 1 & 5 & 2 \\ 0 & 1 & -13 & 2 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 4 & -2 \\ 0 & 1 & 5 & 2 \\ 0 & 0 & -18 & 0 \end{array} \right]$$

$x_3 = 0$
 $x_2 = 2$
 $x_1 + 2x_2 = -2$
 $x_1 = -6$

$$\vec{x} = \begin{bmatrix} -6 \\ 2 \\ 0 \end{bmatrix}$$

- b) (4 points) Without performing any additional row operations, state the solution set for the homogeneous equation $A\mathbf{x} = \mathbf{0}$ where A is the matrix given in part (a) above.

$$\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

9. (2 points) Fill in the blank. At most 3 columns of a 3×5 matrix A form a linearly independent set.

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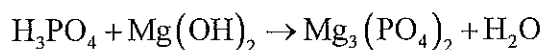
10. (6 points) The following *augmented* matrix in echelon form represents a system of linear equations. For which values of h is the system consistent? Give the solution to the linear system for that h (in parametric vector form).

$$\begin{bmatrix} 1 & 0 & -4 & -1 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & h-3 \end{bmatrix} \quad \checkmark \quad \begin{bmatrix} 1 & 0 & -4 & -1 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad h = 3$$

x_3 free
 $x_2 = 1$
 $x_1 = 4x_3 - 1$

$$\vec{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

11. (9 points) Set up the matrix associated with the homogenous system of equations which could be used to balance the chemical equation. Do not balance the equation.



$$\begin{bmatrix} \text{H} \\ \text{P} \\ \text{O} \\ \text{Mg} \end{bmatrix} \quad \begin{bmatrix} 3 & 2 & 0 & -2 & : & 0 \\ 1 & 0 & -2 & 0 & : & 0 \\ 4 & 2 & -8 & -1 & : & 0 \\ 0 & 1 & -3 & 0 & : & 0 \end{bmatrix}$$

12. (6 points) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation. If $T(x_1, x_2) = \begin{bmatrix} x_2 - 2x_1 \\ 3x_1 + 4x_2 \\ x_2 \end{bmatrix}$ find the matrix of the transformation.

$$A = [T(e_1) \ T(e_2)] = \begin{bmatrix} -2 & 1 \\ 3 & 4 \\ 0 & 1 \end{bmatrix}$$

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13. Consider the vectors $\mathbf{v}_1 = \begin{bmatrix} -1 \\ 0 \\ -3 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

- a) (2 points) Is the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ linearly independent? Yes No (circle one)
- b) (4 points) Find a nonzero vector in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ other than $\mathbf{v}_1$ and $\mathbf{v}_2$, and clearly indicate how you obtained it.

$$2\mathbf{v}_1 = \begin{bmatrix} -2 \\ 0 \\ -6 \end{bmatrix} \text{ OR } \mathbf{v}_1 + \mathbf{v}_2 = \begin{bmatrix} -2 \\ 0 \\ -2 \end{bmatrix} \text{ etc.}$$

- c) (6 points) Give a vector $\mathbf{v}_3$ in $\mathbb{R}^3$ that forms a linearly independent set with $\mathbf{v}_1$ and $\mathbf{v}_2$.

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

14. (9 points) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation. If T is the transformation that first rotates the points clockwise $\pi/4$ and then reflects points through the line $x_1 = -x_2$, find the standard matrix A of the transformation.

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

